

Field Data Review - Sale-level QA summary

Mohawk Valley West Slope demonstration. 798 plots, 8722 trees, BAF 20 on a 300 ft grid across 24 units (1,621 ac). **Sale as a whole** (stratified by unit, area weights): basal area 219 sq ft/ac, sampling error 1.1 % at 95 % confidence against the exhibit 01 standard of 10 % for a tree-measurement sale valued at about \$3,840,213 (77 MBF at an assumed \$50/MBF): **MEETS**. **Strata**: each unit is tested against the 40 % stratum standard for tree-measurement sales (FSH 2409.12 ch. 40, sec. 41.1); the 20-plot column reports the Region 5 practice minimum. QA flags are plots that failed a record check; 7 flagged, 6 of 6 planted errors caught.

Table A. Cruise statistics and QA status by unit (stratum)

Unit	Method	Acres	Plots	BA	CV %	t	SE % (95 %)	Std %	Design	20-plot	QA flags	Status
101	Tractor	97	45	212	13	2.02	3.9	40	meets	ok	7	follow-up
102	Tractor	36	18	219	12	2.11	5.7	40	meets	short	0	accepted
103	Tractor	53	27	204	18	2.06	7.2	40	meets	ok	0	accepted
401	Cable	30	14	221	11	2.16	6.3	40	meets	short	0	accepted
104	Tractor	112	59	207	15	2.00	3.9	40	meets	ok	0	accepted
402	Cable	39	19	214	14	2.10	6.6	40	meets	short	0	accepted
105	Tractor	57	29	222	11	2.05	4.3	40	meets	ok	0	accepted
106	Tractor	89	41	213	20	2.02	6.3	40	meets	ok	0	accepted
107	Tractor	95	46	227	11	2.01	3.3	40	meets	ok	0	accepted
403	Cable	34	20	229	11	2.09	5.2	40	meets	ok	0	accepted
108	Tractor	63	31	222	15	2.04	5.4	40	meets	ok	0	accepted
109	Tractor	73	34	214	25	2.03	8.8	40	meets	ok	0	accepted
404	Cable	51	27	208	19	2.06	7.6	40	meets	ok	0	accepted
110	Tractor	62	31	203	20	2.04	7.5	40	meets	ok	0	accepted
111	Tractor	56	27	222	18	2.06	7.1	40	meets	ok	0	accepted
112	Tractor	80	40	214	16	2.02	5.1	40	meets	ok	0	accepted
113	Tractor	73	36	228	12	2.03	4.1	40	meets	ok	0	accepted
405	Cable	43	20	236	10	2.09	4.7	40	meets	ok	0	accepted
114	Tractor	116	56	212	16	2.00	4.3	40	meets	ok	0	accepted
115	Tractor	84	39	228	12	2.02	4.0	40	meets	ok	0	accepted
116	Tractor	89	44	230	10	2.02	3.1	40	meets	ok	0	accepted
117	Tractor	82	44	223	13	2.02	4.0	40	meets	ok	0	accepted
118	Tractor	70	34	220	18	2.03	6.4	40	meets	ok	0	accepted
406	Cable	35	17	241	9	2.12	4.9	40	meets	short	0	accepted

BA in sq ft/ac; CV = coefficient of variation of plot basal area; t = Student's t at 95 % on n-1 df; SE % = t x standard error / mean.

Unit 101 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 97.4 ac, BAF 20 variable-radius cruise on a 300 ft grid, 45 plots, 478 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	212 sq ft/ac (SE 1.9 %)	Type	Sierran mixed conifer
Trees per acre	123	CWHR size / density	4 / D (cover 83 %)
QMD	17.8 in	SDI	288 (51 % of max 567)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	203 sq ft/ac 19.8 in	BA QMD	9 sq ft/ac 7.6 in
TPA	95	TPA	28
Volume	39.7 MBF/ac (7212 cu ft)	Volume	4.2 green tons/ac (140 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~146 sq ft/ac BA	Sampling error	3.9 % at 95 % (t = 2.02, n = 45)
Species (BA)	PP 70, WF 60, IC 32, DF 29, SP 21	Stratum standard	40 % - MEETS; 45 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	8	15.6	3.6	47	0.3
8-11.9	34	27.7	15.1	290	1.6
12-15.9	74	30.7	32.9	801	4.4
16-19.9	82	21.2	36.4	1074	5.9
20-23.9	68	11.7	30.2	1015	5.6
24-27.9	50	6.0	22.2	814	4.5
28-31.9	52	4.8	23.1	945	5.2
32-35.9	33	2.3	14.7	632	3.5
36-39.9	25	1.4	11.1	527	2.9
40-43.9	18	0.9	8.0	384	2.1
44-47.9	10	0.4	4.4	226	1.2
48-51.9	3	0.1	1.3	73	0.4
52-55.9	9	0.3	4.0	227	1.2
56-59.9	4	0.1	1.8	112	0.6
60-63.9	7	0.2	3.1	185	1.0
All	477	123.4	212.4	7352	40.4

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	158	45.2	70.2	16.9	2404	13.2
WF	134	31.5	59.6	18.6	2061	11.3
IC	71	19.1	31.6	17.4	1097	6.0
DF	66	14.3	29.3	19.4	1077	5.9
SP	47	13.3	20.9	17.0	689	3.8
PPP	1	0.1	0.4	35.1	23	0.1
All	477	123.4	212.4	17.8	7352	40.4

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 101: plots on the 300 ft grid (orange = QA flag)

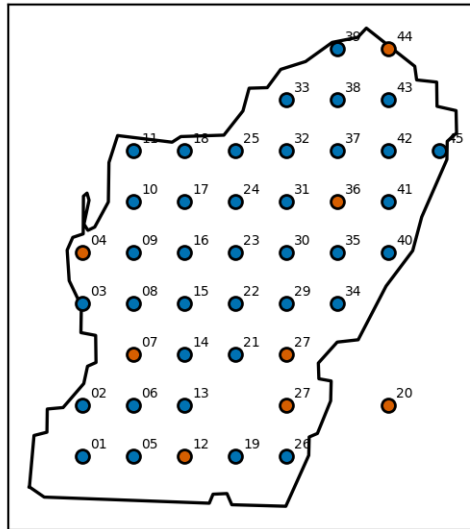


Figure 1. Plot locations with QA status.

QA findings: 7 item(s) need crew follow-up

Plot	Finding
101-04	tree 1: unknown species code 'PPP'
101-07	tree 9: height 32 ft implausible for DBH 27.8
101-12	tree 1: DBH 999.0 out of range
101-20	plot location falls outside its unit boundary (GPS or unit number error)
101-27	duplicate plot id recorded 2 times
101-36	tree 2: height missing
101-44	tree 1: height 15 ft implausible for DBH 34.0

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
101-01	12	240	49	ok
101-02	10	200	63	ok
101-03	6	120	112	ok
101-04	9	180	133	FLAG
101-05	9	180	59	ok
101-06	12	240	60	ok
101-07	13	260	16	FLAG
101-08	11	220	79	ok
101-09	11	220	31	ok
101-10	12	240	107	ok
101-11	9	180	125	ok
101-12	10	200	23	FLAG
101-13	11	220	23	ok
101-14	10	200	55	ok
101-15	10	200	18	ok
101-16	11	220	20	ok
101-17	11	220	15	ok
101-18	10	200	22	ok
101-19	10	200	45	ok
101-20	11	220	111	FLAG
101-21	11	220	4	ok
101-22	10	200	16	ok
101-23	11	220	32	ok
101-24	11	220	53	ok
101-25	11	220	17	ok
101-26	15	300	68	ok
101-27	11	220	44	FLAG
101-27	9	180	60	FLAG
101-29	12	240	26	ok
101-30	11	220	17	ok
101-31	11	220	46	ok
101-32	11	220	18	ok
101-33	9	180	57	ok
101-34	9	180	27	ok
101-35	10	200	27	ok
101-36	12	240	26	FLAG
101-37	11	220	20	ok
101-38	11	220	44	ok
101-39	10	200	44	ok
101-40	12	240	13	ok
101-41	12	240	15	ok
101-42	10	200	12	ok
101-43	9	180	46	ok
101-44	11	220	20	FLAG
101-45	10	200	52	ok

Unit 102 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 36.3 ac, BAF 20 variable-radius cruise on a 300 ft grid, 18 plots, 197 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	219 sq ft/ac (SE 2.7 %)	Type	Sierran mixed conifer
Trees per acre	88	CWHR size / density	4 / D (cover 87 %)
QMD	21.3 in	SDI	285 (50 % of max 574)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	218 sq ft/ac 21.5 in	BA QMD	1 sq ft/ac 9.8 in
TPA	86	TPA	2
Volume	45.2 MBF/ac (8224 cu ft)	Volume	0.6 green tons/ac (19 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~155 sq ft/ac BA	Sampling error	5.7 % at 95 % (t = 2.11, n = 18)
Species (BA)	PP 69, WF 66, DF 33, IC 28, SP 23	Stratum standard	40 % - MEETS; 18 plots vs 20-plot practice: SHORT by 2

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	6	10.7	6.7	142	0.8
12-15.9	18	19.6	20.0	482	2.7
16-19.9	32	20.2	35.6	1061	5.8
20-23.9	40	17.1	44.4	1536	8.4
24-27.9	22	6.7	24.4	921	5.1
28-31.9	30	7.1	33.3	1356	7.5
32-35.9	19	3.4	21.1	936	5.1
36-39.9	8	1.1	8.9	433	2.4
40-43.9	10	1.2	11.1	561	3.1
44-47.9	5	0.5	5.6	314	1.7
48-51.9	1	0.1	1.1	70	0.4
52-55.9	3	0.2	3.3	197	1.1
56-59.9	1	0.1	1.1	77	0.4
60-63.9	2	0.1	2.2	155	0.9
All	197	88.1	218.9	8243	45.3

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	62	28.1	68.9	21.2	2512	13.8
WF	59	26.4	65.6	21.3	2485	13.7
DF	30	15.3	33.3	20.0	1263	6.9
IC	25	9.1	27.8	23.6	1107	6.1
SP	21	9.2	23.3	21.6	876	4.8
All	197	88.1	218.9	21.3	8243	45.3

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 102: plots on the 300 ft grid (orange = QA flag)

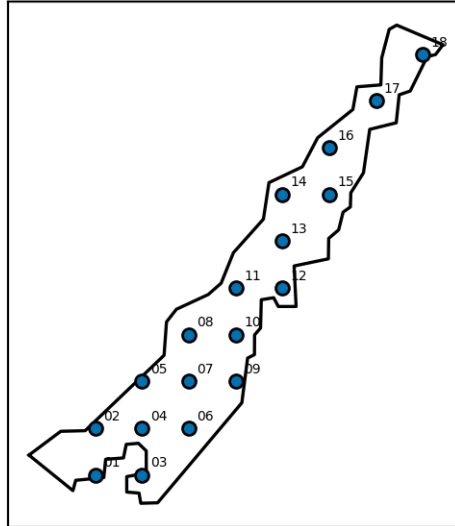


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted; 2 more plots would meet the 20-plot practice

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
102-01	12	240	16	ok
102-02	11	220	154	ok
102-03	9	180	29	ok
102-04	11	220	34	ok
102-05	10	200	14	ok
102-06	12	240	92	ok
102-07	8	160	9	ok
102-08	12	240	54	ok
102-09	10	200	32	ok
102-10	11	220	22	ok
102-11	12	240	22	ok
102-12	12	240	25	ok
102-13	10	200	16	ok
102-14	11	220	537	ok
102-15	10	200	62	ok
102-16	12	240	63	ok
102-17	11	220	37	ok
102-18	13	260	161	ok

Unit 103 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 53.4 ac, BAF 20 variable-radius cruise on a 300 ft grid, 27 plots, 276 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	204 sq ft/ac (SE 3.5 %)	Type	Sierran mixed conifer
Trees per acre	108	CWHR size / density	4 / D (cover 80 %)
QMD	18.6 in	SDI	273 (47 % of max 585)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	196 sq ft/ac 20.6 in	BA QMD	8 sq ft/ac 8.0 in
TPA	85	TPA	23
Volume	40.4 MBF/ac (7341 cu ft)	Volume	3.9 green tons/ac (131 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~153 sq ft/ac BA	Sampling error	7.2 % at 95 % (t = 2.06, n = 27)
Species (BA)	WF 65, PP 58, DF 37, IC 27, SP 18	Stratum standard	40 % - MEETS; 27 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	4	10.6	3.0	45	0.2
8-11.9	19	26.0	14.1	258	1.4
12-15.9	36	24.8	26.7	700	3.8
16-19.9	41	17.7	30.4	878	4.8
20-23.9	46	13.0	34.1	1211	6.7
24-27.9	32	6.5	23.7	898	4.9
28-31.9	26	4.0	19.3	721	4.0
32-35.9	17	2.0	12.6	595	3.3
36-39.9	15	1.4	11.1	539	3.0
40-43.9	12	1.0	8.9	436	2.4
44-47.9	8	0.5	5.9	329	1.8
48-51.9	10	0.5	7.4	412	2.3
52-55.9	2	0.1	1.5	92	0.5
56-59.9	3	0.1	2.2	132	0.7
60-63.9	5	0.2	3.7	226	1.2
All	276	108.4	204.4	7472	41.1

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	88	33.8	65.2	18.8	2533	13.9
PP	78	33.6	57.8	17.8	2018	11.1
DF	50	18.3	37.0	19.2	1363	7.5
IC	36	13.6	26.7	19.0	956	5.3
SP	24	9.1	17.8	18.9	601	3.3
All	276	108.4	204.4	18.6	7472	41.1

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 103: plots on the 300 ft grid (orange = QA flag)

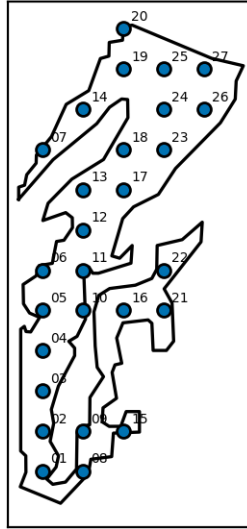


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
103-01	9	180	9	ok
103-02	11	220	174	ok
103-03	14	280	103	ok
103-04	10	200	157	ok
103-05	12	240	109	ok
103-06	8	160	132	ok
103-07	12	240	138	ok
103-08	10	200	127	ok
103-09	7	140	103	ok
103-10	9	180	159	ok
103-11	11	220	141	ok
103-12	8	160	95	ok
103-13	10	200	81	ok
103-14	13	260	113	ok
103-15	11	220	101	ok
103-16	9	180	70	ok
103-17	10	200	55	ok
103-18	12	240	56	ok
103-19	9	180	76	ok
103-20	7	140	62	ok
103-21	7	140	26	ok
103-22	10	200	127	ok
103-23	11	220	70	ok
103-24	12	240	63	ok
103-25	12	240	67	ok
103-26	12	240	68	ok
103-27	10	200	34	ok

Unit 401 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 29.5 ac, BAF 20 variable-radius cruise on a 300 ft grid, 14 plots, 155 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	221 sq ft/ac (SE 2.9 %)	Type	Sierran mixed conifer
Trees per acre	101	CWHR size / density	4 / D (cover 94 %)
QMD	20.0 in	SDI	290 (49 % of max 591)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	217 sq ft/ac 20.8 in	BA QMD	4 sq ft/ac 9.5 in
TPA	92	TPA	9
Volume	45.9 MBF/ac (8345 cu ft)	Volume	2.4 green tons/ac (82 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~158 sq ft/ac BA	Sampling error	6.3 % at 95 % (t = 2.16, n = 14)
Species (BA)	WF 79, PP 66, DF 33, IC 24, SP 20	Stratum standard	40 % - MEETS; 14 plots vs 20-plot practice: SHORT by 6

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	9	21.0	12.9	231	1.3
12-15.9	27	35.9	38.6	916	5.0
16-19.9	19	15.2	27.1	795	4.4
20-23.9	15	8.4	21.4	691	3.8
24-27.9	19	7.6	27.1	1113	6.1
28-31.9	14	4.1	20.0	802	4.4
32-35.9	23	5.2	32.9	1533	8.4
36-39.9	8	1.5	11.4	587	3.2
40-43.9	1	0.1	1.4	68	0.4
44-47.9	5	0.6	7.1	378	2.1
48-51.9	10	1.1	14.3	865	4.8
52-55.9	2	0.2	2.9	187	1.0
56-59.9	1	0.1	1.4	82	0.5
60-63.9	2	0.1	2.9	179	1.0
All	155	101.1	221.4	8427	46.3

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	55	31.7	78.6	21.3	3021	16.6
PP	46	30.6	65.7	19.8	2585	14.2
DF	23	13.8	32.9	20.9	1339	7.4
IC	17	12.9	24.3	18.6	774	4.3
SP	14	12.0	20.0	17.5	708	3.9
All	155	101.1	221.4	20.0	8427	46.3

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

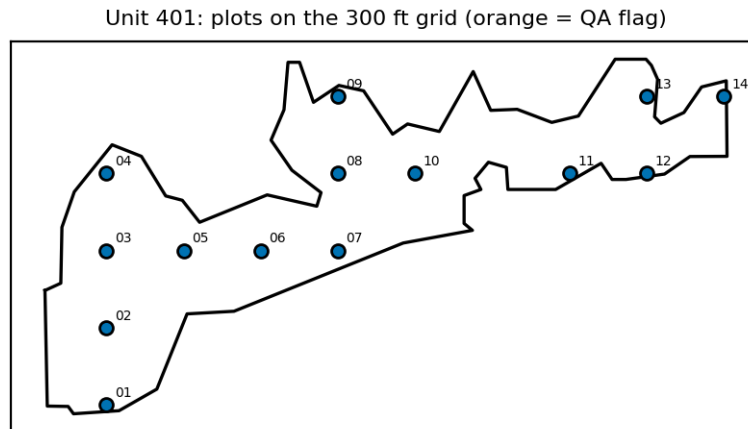


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted; 6 more plots would meet the 20-plot practice

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
401-01	12	240	260	ok
401-02	10	200	219	ok
401-03	11	220	116	ok
401-04	13	260	182	ok
401-05	12	240	126	ok
401-06	10	200	102	ok
401-07	11	220	198	ok
401-08	13	260	127	ok
401-09	10	200	183	ok
401-10	11	220	178	ok
401-11	11	220	107	ok
401-12	9	180	195	ok
401-13	10	200	124	ok
401-14	12	240	163	ok

Unit 104 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 111.5 ac, BAF 20 variable-radius cruise on a 300 ft grid, 59 plots, 611 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	207 sq ft/ac (SE 2.0 %)	Type	Sierran mixed conifer
Trees per acre	165	CWHR size / density	4 / D (cover 85 %)
QMD	15.2 in	SDI	300 (52 % of max 572)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	186 sq ft/ac 18.2 in	BA QMD	21 sq ft/ac 7.8 in
TPA	103	TPA	62
Volume	32.0 MBF/ac (5822 cu ft)	Volume	10.2 green tons/ac (341 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~138 sq ft/ac BA	Sampling error	3.9 % at 95 % (t = 2.00, n = 59)
Species (BA)	PP 66, WF 61, IC 32, DF 29, SP 20	Stratum standard	40 % - MEETS; 59 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	21	31.5	7.1	106	0.6
8-11.9	79	50.4	26.8	480	2.6
12-15.9	108	34.9	36.6	819	4.5
16-19.9	121	23.9	41.0	1133	6.2
20-23.9	81	10.5	27.5	860	4.7
24-27.9	78	7.4	26.4	917	5.0
28-31.9	38	2.7	12.9	511	2.8
32-35.9	30	1.6	10.2	440	2.4
36-39.9	24	1.1	8.1	370	2.0
40-43.9	12	0.4	4.1	197	1.1
44-47.9	7	0.2	2.4	124	0.7
48-51.9	6	0.2	2.0	104	0.6
52-55.9	2	0.0	0.7	35	0.2
56-59.9	2	0.0	0.7	34	0.2
60-63.9	2	0.0	0.7	33	0.2
All	611	164.8	207.1	6163	33.9

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	195	49.2	66.1	15.7	2026	11.1
WF	179	51.6	60.7	14.7	1815	10.0
IC	93	24.7	31.5	15.3	902	5.0
DF	86	20.6	29.2	16.1	871	4.8
SP	58	18.5	19.7	14.0	549	3.0
All	611	164.8	207.1	15.2	6163	33.9

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 104: plots on the 300 ft grid (orange = QA flag)

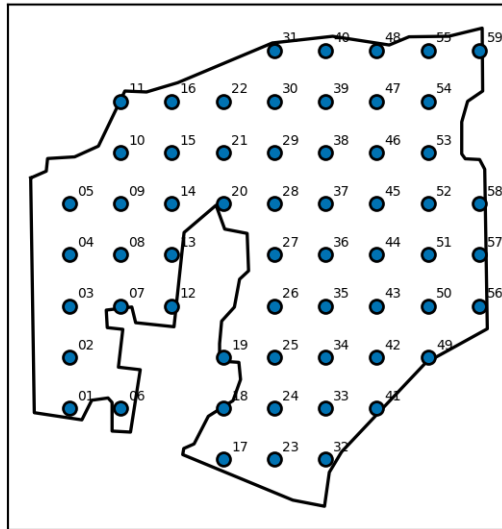


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
104-01	8	160	57	ok
104-02	12	240	36	ok
104-03	12	240	21	ok
104-04	11	220	35	ok
104-05	6	120	14	ok
104-06	10	200	100	ok
104-07	8	160	96	ok
104-08	11	220	34	ok
104-09	8	160	28	ok
104-10	9	180	33	ok
104-11	9	180	94	ok
104-12	10	200	105	ok
104-13	11	220	86	ok
104-14	10	200	21	ok
104-15	9	180	42	ok
104-16	8	160	37	ok
104-17	11	220	63	ok
104-18	12	240	69	ok
104-19	9	180	167	ok
104-20	12	240	148	ok
104-21	9	180	19	ok
104-22	11	220	30	ok
104-23	11	220	84	ok
104-24	10	200	83	ok
104-25	12	240	73	ok
104-26	10	200	134	ok
104-27	13	260	60	ok
104-28	11	220	25	ok
104-29	8	160	38	ok
104-30	11	220	17	ok
104-31	11	220	54	ok
104-32	11	220	11	ok
104-33	11	220	46	ok
104-34	8	160	50	ok
104-35	12	240	38	ok
104-36	13	260	34	ok
104-37	11	220	107	ok
104-38	11	220	60	ok
104-39	10	200	32	ok
104-40	11	220	36	ok
104-41	11	220	35	ok
104-42	12	240	37	ok
104-43	10	200	31	ok
104-44	12	240	10	ok
104-45	11	220	41	ok
104-46	9	180	123	ok
104-47	12	240	17	ok
104-48	9	180	55	ok
104-49	10	200	77	ok
104-50	6	120	40	ok
104-51	12	240	26	ok

Plot	Trees in	BA sq ft/ac	Slope %	Status
104-52	12	240	9	ok
104-53	11	220	94	ok
104-54	10	200	21	ok
104-55	12	240	13	ok
104-56	11	220	47	ok
104-57	11	220	28	ok
104-58	10	200	43	ok
104-59	9	180	15	ok

Unit 402 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 38.9 ac, BAF 20 variable-radius cruise on a 300 ft grid, 19 plots, 203 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (SE 3.1 %)	Type	Sierran mixed conifer
Trees per acre	64	CWHR size / density	5 / D (cover 90 %)
QMD	24.8 in	SDI	261 (45 % of max 576)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	214 sq ft/ac 24.8 in	BA QMD	0 sq ft/ac 0.0 in
TPA	64	TPA	0
Volume	53.4 MBF/ac (9708 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~165 sq ft/ac BA	Sampling error	6.6 % at 95 % (t = 2.10, n = 19)
Species (BA)	PP 67, WF 65, DF 36, IC 27, SP 18	Stratum standard	40 % - MEETS; 19 plots vs 20-plot practice: SHORT by 1

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
12-15.9	14	13.1	14.7	440	2.4
16-19.9	30	17.5	31.6	1046	5.8
20-23.9	33	13.0	34.7	1292	7.1
24-27.9	28	8.0	29.5	1253	6.9
28-31.9	17	3.6	17.9	811	4.5
32-35.9	12	2.0	12.6	625	3.4
36-39.9	20	2.7	21.1	1073	5.9
40-43.9	10	1.1	10.5	561	3.1
44-47.9	15	1.4	15.8	955	5.3
48-51.9	5	0.4	5.3	312	1.7
52-55.9	3	0.2	3.2	202	1.1
56-59.9	4	0.2	4.2	290	1.6
60-63.9	12	0.6	12.6	848	4.7
All	203	63.8	213.7	9708	53.4

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	64	16.8	67.4	27.2	3216	17.7
WF	62	20.7	65.3	24.1	2901	16.0
DF	34	12.0	35.8	23.4	1576	8.7
IC	26	9.5	27.4	23.0	1212	6.7
SP	17	4.8	17.9	26.1	803	4.4
All	203	63.8	213.7	24.8	9708	53.4

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 402: plots on the 300 ft grid (orange = QA flag)

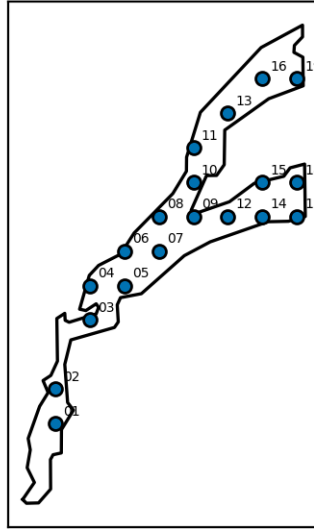


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted; 1 more plots would meet the 20-plot practice

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
402-01	7	140	116	ok
402-02	12	240	236	ok
402-03	10	200	208	ok
402-04	12	240	209	ok
402-05	12	240	227	ok
402-06	9	180	57	ok
402-07	11	220	198	ok
402-08	10	200	200	ok
402-09	10	200	88	ok
402-10	12	240	169	ok
402-11	9	180	141	ok
402-12	11	220	255	ok
402-13	10	200	38	ok
402-14	11	220	178	ok
402-15	10	200	202	ok
402-16	12	240	271	ok
402-17	12	240	203	ok
402-18	13	260	141	ok
402-19	10	200	154	ok

Unit 105 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 57.4 ac, BAF 20 variable-radius cruise on a 300 ft grid, 29 plots, 322 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (SE 2.1 %)	Type	Sierran mixed conifer
Trees per acre	71	CWHR size / density	4 / D (cover 93 %)
QMD	24.0 in	SDI	274 (48 % of max 565)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	221 sq ft/ac 24.8 in	BA QMD	1 sq ft/ac 7.2 in
TPA	66	TPA	5
Volume	51.7 MBF/ac (9397 cu ft)	Volume	0.8 green tons/ac (27 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~161 sq ft/ac BA	Sampling error	4.3 % at 95 % (t = 2.05, n = 29)
Species (BA)	PP 72, WF 60, DF 34, IC 32, SP 23	Stratum standard	40 % - MEETS; 29 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	3.2	0.7	10	0.1
8-11.9	5	6.1	3.4	77	0.4
12-15.9	16	9.6	11.0	314	1.7
16-19.9	36	13.9	24.8	819	4.5
20-23.9	50	13.2	34.5	1256	6.9
24-27.9	46	8.5	31.7	1243	6.8
28-31.9	48	6.8	33.1	1429	7.9
32-35.9	43	4.8	29.7	1327	7.3
36-39.9	18	1.6	12.4	645	3.5
40-43.9	17	1.2	11.7	603	3.3
44-47.9	10	0.6	6.9	372	2.0
48-51.9	11	0.6	7.6	435	2.4
52-55.9	4	0.2	2.8	159	0.9
56-59.9	6	0.2	4.1	259	1.4
60-63.9	11	0.4	7.6	476	2.6
All	322	70.9	222.1	9424	51.8

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	105	21.5	72.4	24.9	3147	17.3
WF	87	22.1	60.0	22.3	2455	13.5
DF	50	12.0	34.5	23.0	1413	7.8
IC	46	7.6	31.7	27.6	1383	7.6
SP	34	7.7	23.4	23.6	1027	5.6
All	322	70.9	222.1	24.0	9424	51.8

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 105: plots on the 300 ft grid (orange = QA flag)

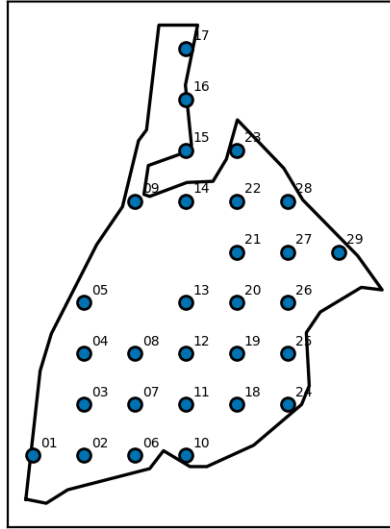


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
105-01	11	220	33	ok
105-02	12	240	55	ok
105-03	12	240	29	ok
105-04	14	280	113	ok
105-05	10	200	78	ok
105-06	10	200	36	ok
105-07	11	220	10	ok
105-08	12	240	87	ok
105-09	11	220	102	ok
105-10	13	260	54	ok
105-11	11	220	67	ok
105-12	10	200	35	ok
105-13	10	200	65	ok
105-14	10	200	61	ok
105-15	9	180	73	ok
105-16	10	200	138	ok
105-17	11	220	129	ok
105-18	11	220	68	ok
105-19	8	160	75	ok
105-20	12	240	113	ok
105-21	11	220	111	ok
105-22	12	240	64	ok
105-23	13	260	84	ok
105-24	12	240	47	ok
105-25	10	200	62	ok
105-26	11	220	134	ok
105-27	12	240	57	ok
105-28	12	240	52	ok
105-29	11	220	60	ok

Unit 106 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 88.6 ac, BAF 20 variable-radius cruise on a 300 ft grid, 41 plots, 437 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	213 sq ft/ac (SE 3.1 %)	Type	Sierran mixed conifer
Trees per acre	95	CWHR size / density	4 / D (cover 85 %)
QMD	20.3 in	SDI	272 (47 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	207 sq ft/ac 22.9 in	BA QMD	6 sq ft/ac 7.0 in
TPA	73	TPA	22
Volume	47.1 MBF/ac (8568 cu ft)	Volume	2.4 green tons/ac (80 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~159 sq ft/ac BA	Sampling error	6.3 % at 95 % (t = 2.02, n = 41)
Species (BA)	WF 68, PP 65, IC 32, DF 29, SP 19	Stratum standard	40 % - MEETS; 41 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	9	18.8	4.4	56	0.3
8-11.9	12	10.2	5.9	112	0.6
12-15.9	39	18.5	19.0	512	2.8
16-19.9	55	15.2	26.8	860	4.7
20-23.9	49	9.3	23.9	837	4.6
24-27.9	64	8.6	31.2	1250	6.9
28-31.9	53	5.3	25.9	1133	6.2
32-35.9	51	4.0	24.9	1125	6.2
36-39.9	28	1.8	13.7	687	3.8
40-43.9	29	1.5	14.1	710	3.9
44-47.9	11	0.5	5.4	302	1.7
48-51.9	15	0.5	7.3	428	2.4
52-55.9	6	0.2	2.9	163	0.9
56-59.9	5	0.1	2.4	138	0.8
60-63.9	11	0.3	5.4	335	1.8
All	437	94.7	213.2	8648	47.6

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	139	29.2	67.8	20.6	2777	15.3
PP	134	27.8	65.4	20.7	2700	14.8
IC	66	11.9	32.2	22.2	1314	7.2
DF	59	13.9	28.8	19.5	1134	6.2
SP	39	11.8	19.0	17.2	723	4.0
All	437	94.7	213.2	20.3	8648	47.6

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 106: plots on the 300 ft grid (orange = QA flag)

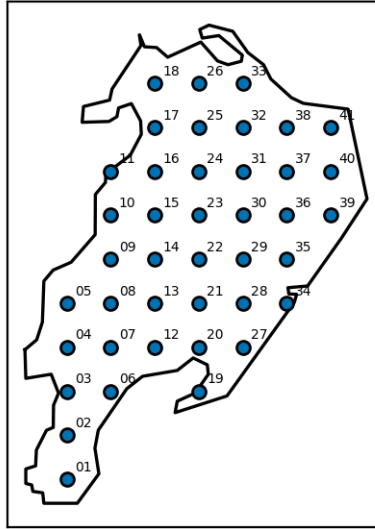


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
106-01	9	180	133	ok
106-02	7	140	34	ok
106-03	12	240	77	ok
106-04	13	260	32	ok
106-05	13	260	31	ok
106-06	11	220	44	ok
106-07	12	240	82	ok
106-08	8	160	17	ok
106-09	10	200	26	ok
106-10	14	280	17	ok
106-11	13	260	32	ok
106-12	8	160	177	ok
106-13	7	140	26	ok
106-14	10	200	9	ok
106-15	13	260	39	ok
106-16	11	220	34	ok
106-17	9	180	142	ok
106-18	11	220	64	ok
106-19	12	240	79	ok
106-20	12	240	90	ok
106-21	7	140	19	ok
106-22	12	240	67	ok
106-23	12	240	15	ok
106-24	11	220	26	ok
106-25	10	200	32	ok
106-26	13	260	81	ok
106-27	11	220	100	ok
106-28	10	200	96	ok
106-29	10	200	21	ok
106-30	10	200	62	ok
106-31	11	220	53	ok
106-32	11	220	10	ok
106-33	12	240	36	ok
106-34	11	220	96	ok
106-35	12	240	15	ok
106-36	12	240	13	ok
106-37	11	220	37	ok
106-38	11	220	36	ok
106-39	3	60	14	ok
106-40	10	200	27	ok
106-41	12	240	13	ok

Unit 107 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 95.3 ac, BAF 20 variable-radius cruise on a 300 ft grid, 46 plots, 522 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	227 sq ft/ac (SE 1.6 %)	Type	Sierran mixed conifer
Trees per acre	72	CWHR size / density	4 / D (cover 93 %)
QMD	24.0 in	SDI	280 (48 % of max 584)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	226 sq ft/ac 24.4 in	BA QMD	1 sq ft/ac 8.9 in
TPA	69	TPA	3
Volume	53.1 MBF/ac (9648 cu ft)	Volume	0.9 green tons/ac (30 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~165 sq ft/ac BA	Sampling error	3.3 % at 95 % (t = 2.01, n = 46)
Species (BA)	WF 71, PP 65, IC 37, DF 33, SP 21	Stratum standard	40 % - MEETS; 46 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	7	5.6	3.0	76	0.4
12-15.9	33	13.0	14.3	401	2.2
16-19.9	62	15.5	27.0	886	4.9
20-23.9	79	13.5	34.3	1286	7.1
24-27.9	82	9.9	35.7	1439	7.9
28-31.9	61	5.5	26.5	1100	6.0
32-35.9	54	3.8	23.5	1071	5.9
36-39.9	42	2.4	18.3	891	4.9
40-43.9	30	1.4	13.0	676	3.7
44-47.9	18	0.7	7.8	429	2.4
48-51.9	18	0.6	7.8	443	2.4
52-55.9	14	0.4	6.1	374	2.1
56-59.9	8	0.2	3.5	204	1.1
60-63.9	14	0.3	6.1	401	2.2
All	522	72.5	227.0	9677	53.2

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	164	22.5	71.3	24.1	3063	16.8
PP	149	20.1	64.8	24.3	2729	15.0
IC	85	13.1	37.0	22.8	1576	8.7
DF	76	10.2	33.0	24.4	1383	7.6
SP	48	6.5	20.9	24.2	926	5.1
All	522	72.5	227.0	24.0	9677	53.2

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 107: plots on the 300 ft grid (orange = QA flag)

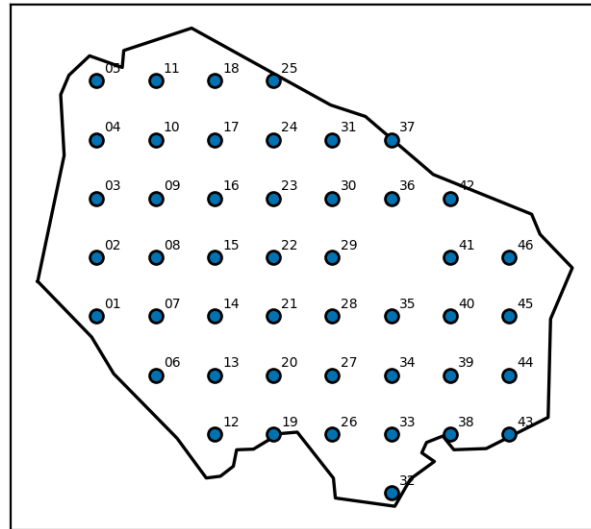


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
107-01	13	260	28	ok
107-02	12	240	18	ok
107-03	13	260	110	ok
107-04	11	220	31	ok
107-05	8	160	62	ok
107-06	11	220	20	ok
107-07	11	220	43	ok
107-08	13	260	7	ok
107-09	12	240	55	ok
107-10	10	200	56	ok
107-11	12	240	61	ok
107-12	11	220	52	ok
107-13	10	200	64	ok
107-14	13	260	127	ok
107-15	13	260	59	ok
107-16	11	220	55	ok
107-17	9	180	125	ok
107-18	12	240	47	ok
107-19	12	240	88	ok
107-20	10	200	87	ok
107-21	12	240	60	ok
107-22	12	240	50	ok
107-23	11	220	103	ok
107-24	12	240	28	ok
107-25	10	200	97	ok
107-26	8	160	84	ok
107-27	10	200	50	ok
107-28	11	220	60	ok
107-29	12	240	149	ok
107-30	11	220	32	ok
107-31	11	220	66	ok
107-32	12	240	78	ok
107-33	10	200	93	ok
107-34	11	220	24	ok
107-35	11	220	58	ok
107-36	11	220	24	ok
107-37	12	240	67	ok
107-38	12	240	108	ok
107-39	12	240	103	ok
107-40	12	240	61	ok
107-41	12	240	59	ok
107-42	13	260	107	ok
107-43	14	280	68	ok
107-44	10	200	76	ok
107-45	11	220	46	ok
107-46	12	240	60	ok

Unit 403 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 34.0 ac, BAF 20 variable-radius cruise on a 300 ft grid, 20 plots, 229 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	229 sq ft/ac (SE 2.5 %)	Type	Sierran mixed conifer
Trees per acre	63	CWHR size / density	5 / D (cover 95 %)
QMD	25.9 in	SDI	277 (49 % of max 562)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	229 sq ft/ac 25.9 in	BA QMD	0 sq ft/ac 0.0 in
TPA	63	TPA	0
Volume	55.5 MBF/ac (10093 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~163 sq ft/ac BA	Sampling error	5.2 % at 95 % (t = 2.09, n = 20)
Species (BA)	PP 68, WF 53, DF 40, IC 38, SP 30	Stratum standard	40 % - MEETS; 20 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	1	1.5	1.0	25	0.1
12-15.9	9	8.4	9.0	278	1.5
16-19.9	23	13.2	23.0	804	4.4
20-23.9	34	12.9	34.0	1278	7.0
24-27.9	33	8.9	33.0	1357	7.5
28-31.9	29	6.0	29.0	1241	6.8
32-35.9	28	4.4	28.0	1333	7.3
36-39.9	25	3.2	25.0	1219	6.7
40-43.9	18	1.9	18.0	931	5.1
44-47.9	12	1.1	12.0	650	3.6
48-51.9	4	0.3	4.0	210	1.2
52-55.9	4	0.3	4.0	225	1.2
56-59.9	5	0.3	5.0	299	1.6
60-63.9	4	0.2	4.0	244	1.3
All	229	62.6	229.0	10093	55.5

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	68	19.1	68.0	25.5	3057	16.8
WF	53	14.9	53.0	25.5	2320	12.8
DF	40	10.0	40.0	27.1	1770	9.7
IC	38	9.2	38.0	27.6	1682	9.3
SP	30	9.4	30.0	24.2	1264	7.0
All	229	62.6	229.0	25.9	10093	55.5

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

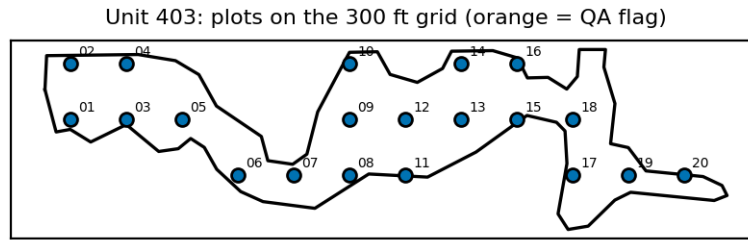


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
403-01	12	240	140	ok
403-02	13	260	241	ok
403-03	13	260	131	ok
403-04	13	260	145	ok
403-05	9	180	141	ok
403-06	10	200	127	ok
403-07	10	200	110	ok
403-08	10	200	119	ok
403-09	12	240	155	ok
403-10	13	260	125	ok
403-11	12	240	107	ok
403-12	12	240	150	ok
403-13	11	220	120	ok
403-14	11	220	134	ok
403-15	10	200	150	ok
403-16	12	240	130	ok
403-17	13	260	144	ok
403-18	11	220	154	ok
403-19	10	200	157	ok
403-20	12	240	81	ok

Unit 108 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 63.3 ac, BAF 20 variable-radius cruise on a 300 ft grid, 31 plots, 344 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (SE 2.6 %)	Type	Sierran mixed conifer
Trees per acre	68	CWHR size / density	5 / D (cover 91 %)
QMD	24.5 in	SDI	274 (47 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	222 sq ft/ac 24.5 in	BA QMD	0 sq ft/ac 0.0 in
TPA	68	TPA	0
Volume	52.3 MBF/ac (9501 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~164 sq ft/ac BA	Sampling error	5.4 % at 95 % (t = 2.04, n = 31)
Species (BA)	WF 68, PP 66, IC 36, DF 29, SP 22	Stratum standard	40 % - MEETS; 31 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	4	3.6	2.6	81	0.4
12-15.9	22	13.2	14.2	425	2.3
16-19.9	33	12.0	21.3	744	4.1
20-23.9	53	12.7	34.2	1310	7.2
24-27.9	60	10.7	38.7	1582	8.7
28-31.9	44	5.9	28.4	1227	6.7
32-35.9	39	4.0	25.2	1152	6.3
36-39.9	35	2.9	22.6	1100	6.1
40-43.9	26	1.7	16.8	860	4.7
44-47.9	6	0.3	3.9	188	1.0
48-51.9	7	0.3	4.5	250	1.4
52-55.9	6	0.2	3.9	230	1.3
56-59.9	5	0.2	3.2	186	1.0
60-63.9	4	0.1	2.6	166	0.9
All	344	68.1	221.9	9501	52.3

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	106	22.5	68.4	23.6	2863	15.7
PP	103	19.0	66.5	25.3	2891	15.9
IC	56	12.9	36.1	22.7	1471	8.1
DF	45	7.0	29.0	27.6	1290	7.1
SP	34	6.7	21.9	24.4	986	5.4
All	344	68.1	221.9	24.5	9501	52.3

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 108: plots on the 300 ft grid (orange = QA flag)

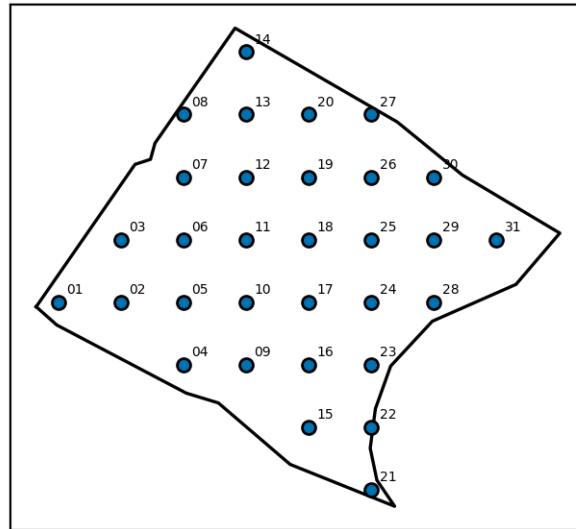


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
108-01	10	200	32	ok
108-02	11	220	71	ok
108-03	9	180	69	ok
108-04	10	200	65	ok
108-05	10	200	37	ok
108-06	12	240	69	ok
108-07	11	220	81	ok
108-08	11	220	198	ok
108-09	11	220	82	ok
108-10	12	240	51	ok
108-11	11	220	69	ok
108-12	14	280	71	ok
108-13	6	120	99	ok
108-14	13	260	16	ok
108-15	14	280	64	ok
108-16	10	200	64	ok
108-17	11	220	29	ok
108-18	11	220	14	ok
108-19	9	180	133	ok
108-20	12	240	197	ok
108-21	10	200	74	ok
108-22	12	240	40	ok
108-23	11	220	81	ok
108-24	12	240	37	ok
108-25	10	200	43	ok
108-26	13	260	48	ok
108-27	13	260	243	ok
108-28	11	220	16	ok
108-29	11	220	50	ok
108-30	10	200	67	ok
108-31	13	260	33	ok

Unit 109 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 72.9 ac, BAF 20 variable-radius cruise on a 300 ft grid, 34 plots, 364 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (SE 4.3 %)	Type	Sierran mixed conifer
Trees per acre	65	CWHR size / density	5 / D (cover 88 %)
QMD	24.6 in	SDI	263 (44 % of max 592)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	213 sq ft/ac 25.1 in	BA QMD	1 sq ft/ac 8.8 in
TPA	62	TPA	3
Volume	51.1 MBF/ac (9290 cu ft)	Volume	0.9 green tons/ac (29 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~169 sq ft/ac BA	Sampling error	8.8 % at 95 % (t = 2.03, n = 34)
Species (BA)	WF 70, PP 55, DF 36, IC 32, SP 21	Stratum standard	40 % - MEETS; 34 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	4	4.3	2.4	64	0.4
12-15.9	17	9.4	10.0	292	1.6
16-19.9	40	12.7	23.5	807	4.4
20-23.9	68	15.6	40.0	1506	8.3
24-27.9	44	7.1	25.9	1025	5.6
28-31.9	49	6.0	28.8	1271	7.0
32-35.9	39	3.7	22.9	1087	6.0
36-39.9	31	2.3	18.2	922	5.1
40-43.9	25	1.5	14.7	767	4.2
44-47.9	13	0.7	7.6	408	2.2
48-51.9	9	0.4	5.3	293	1.6
52-55.9	9	0.3	5.3	297	1.6
56-59.9	6	0.2	3.5	210	1.2
60-63.9	10	0.3	5.9	371	2.0
All	364	64.7	214.1	9319	51.3

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	119	21.1	70.0	24.7	3037	16.7
PP	94	16.0	55.3	25.2	2393	13.2
DF	62	11.2	36.5	24.4	1576	8.7
IC	54	10.5	31.8	23.5	1377	7.6
SP	35	5.9	20.6	25.3	935	5.1
All	364	64.7	214.1	24.6	9319	51.3

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 109: plots on the 300 ft grid (orange = QA flag)

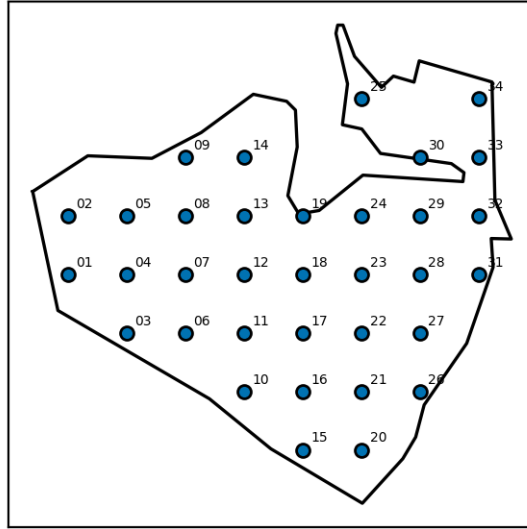


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
109-01	15	300	27	ok
109-02	10	200	42	ok
109-03	10	200	102	ok
109-04	11	220	16	ok
109-05	9	180	22	ok
109-06	12	240	104	ok
109-07	3	60	18	ok
109-08	12	240	21	ok
109-09	12	240	52	ok
109-10	11	220	69	ok
109-11	13	260	55	ok
109-12	11	220	31	ok
109-13	15	300	16	ok
109-14	13	260	25	ok
109-15	10	200	62	ok
109-16	10	200	32	ok
109-17	10	200	47	ok
109-18	10	200	24	ok
109-19	7	140	229	ok
109-20	12	240	39	ok
109-21	12	240	25	ok
109-22	9	180	8	ok
109-23	11	220	17	ok
109-24	13	260	86	ok
109-25	10	200	63	ok
109-26	10	200	75	ok
109-27	10	200	52	ok
109-28	14	280	42	ok
109-29	11	220	59	ok
109-30	11	220	81	ok
109-31	13	260	147	ok
109-32	2	40	185	ok
109-33	12	240	111	ok
109-34	10	200	31	ok

Unit 404 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 51.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 27 plots, 281 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	208 sq ft/ac (SE 3.7 %)	Type	Sierran mixed conifer
Trees per acre	61	CWHR size / density	5 / D (cover 87 %)
QMD	25.0 in	SDI	252 (43 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	207 sq ft/ac 25.7 in	BA QMD	1 sq ft/ac 5.9 in
TPA	57	TPA	4
Volume	51.9 MBF/ac (9437 cu ft)	Volume	0.3 green tons/ac (8 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~168 sq ft/ac BA	Sampling error	7.6 % at 95 % (t = 2.06, n = 27)
Species (BA)	WF 67, PP 64, DF 30, IC 27, SP 20	Stratum standard	40 % - MEETS; 27 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	3.9	0.7	8	0.0
8-11.9	4	4.4	3.0	58	0.3
12-15.9	10	6.5	7.4	219	1.2
16-19.9	27	11.5	20.0	686	3.8
20-23.9	36	10.2	26.7	983	5.4
24-27.9	43	8.9	31.9	1272	7.0
28-31.9	38	5.8	28.1	1303	7.2
32-35.9	37	4.3	27.4	1294	7.1
36-39.9	22	2.1	16.3	837	4.6
40-43.9	13	1.0	9.6	497	2.7
44-47.9	17	1.1	12.6	715	3.9
48-51.9	17	0.9	12.6	787	4.3
52-55.9	3	0.1	2.2	134	0.7
56-59.9	3	0.1	2.2	148	0.8
60-63.9	10	0.4	7.4	504	2.8
All	281	61.3	208.1	9446	52.0

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	90	23.7	66.7	22.7	2957	16.3
PP	87	16.6	64.4	26.6	2940	16.2
DF	41	7.5	30.4	27.2	1423	7.8
IC	36	6.7	26.7	27.0	1182	6.5
SP	27	6.7	20.0	23.4	944	5.2
All	281	61.3	208.1	25.0	9446	52.0

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 404: plots on the 300 ft grid (orange = QA flag)

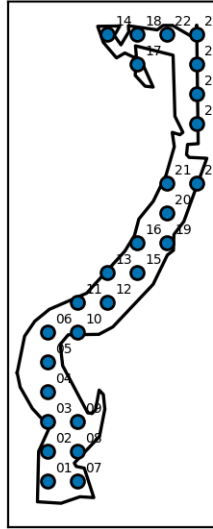


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
404-01	11	220	194	ok
404-02	11	220	155	ok
404-03	13	260	174	ok
404-04	10	200	222	ok
404-05	11	220	181	ok
404-06	11	220	173	ok
404-07	12	240	159	ok
404-08	9	180	129	ok
404-09	13	260	182	ok
404-10	6	120	212	ok
404-11	12	240	163	ok
404-12	11	220	238	ok
404-13	12	240	148	ok
404-14	9	180	154	ok
404-15	8	160	189	ok
404-16	10	200	195	ok
404-17	11	220	88	ok
404-18	10	200	165	ok
404-19	4	80	16	ok
404-20	10	200	391	ok
404-21	11	220	217	ok
404-22	10	200	168	ok
404-23	10	200	152	ok
404-24	10	200	145	ok
404-25	11	220	236	ok
404-26	12	240	213	ok
404-27	13	260	98	ok

Unit 110 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 62.0 ac, BAF 20 variable-radius cruise on a 300 ft grid, 31 plots, 315 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	203 sq ft/ac (SE 3.7 %)	Type	Sierran mixed conifer
Trees per acre	137	CWHR size / density	4 / D (cover 82 %)
QMD	16.5 in	SDI	287 (50 % of max 572)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	194 sq ft/ac 18.4 in	BA QMD	10 sq ft/ac 7.5 in
TPA	105	TPA	32
Volume	34.1 MBF/ac (6200 cu ft)	Volume	4.5 green tons/ac (150 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~142 sq ft/ac BA	Sampling error	7.5 % at 95 % (t = 2.04, n = 31)
Species (BA)	PP 65, WF 60, IC 28, DF 26, SP 25	Stratum standard	40 % - MEETS; 31 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	6	18.9	3.9	50	0.3
8-11.9	27	31.5	17.4	340	1.9
12-15.9	60	36.7	38.7	916	5.0
16-19.9	68	25.4	43.9	1255	6.9
20-23.9	46	11.5	29.7	888	4.9
24-27.9	30	5.4	19.4	701	3.9
28-31.9	32	4.3	20.6	798	4.4
32-35.9	15	1.6	9.7	396	2.2
36-39.9	10	0.8	6.5	291	1.6
40-43.9	5	0.3	3.2	156	0.9
44-47.9	5	0.3	3.2	169	0.9
48-51.9	3	0.1	1.9	108	0.6
52-55.9	3	0.1	1.9	106	0.6
56-59.9	2	0.1	1.3	68	0.4
60-63.9	3	0.1	1.9	108	0.6
All	315	137.0	203.2	6350	34.9

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	101	46.5	65.2	16.0	1987	10.9
WF	93	36.5	60.0	17.4	1893	10.4
IC	43	21.4	27.7	15.4	860	4.7
DF	40	16.1	25.8	17.1	855	4.7
SP	38	16.5	24.5	16.5	755	4.2
All	315	137.0	203.2	16.5	6350	34.9

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 110: plots on the 300 ft grid (orange = QA flag)

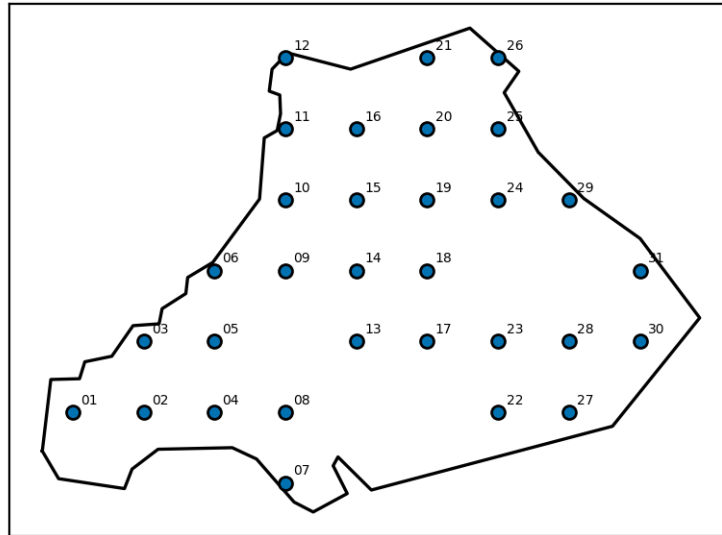


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
110-01	6	120	68	ok
110-02	13	260	89	ok
110-03	11	220	50	ok
110-04	8	160	54	ok
110-05	7	140	25	ok
110-06	10	200	82	ok
110-07	13	260	131	ok
110-08	13	260	85	ok
110-09	12	240	57	ok
110-10	6	120	114	ok
110-11	10	200	141	ok
110-12	7	140	74	ok
110-13	9	180	117	ok
110-14	11	220	98	ok
110-15	10	200	83	ok
110-16	11	220	145	ok
110-17	12	240	76	ok
110-18	10	200	121	ok
110-19	10	200	55	ok
110-20	12	240	116	ok
110-21	13	260	30	ok
110-22	11	220	37	ok
110-23	11	220	48	ok
110-24	10	200	85	ok
110-25	10	200	59	ok
110-26	8	160	81	ok
110-27	12	240	110	ok
110-28	10	200	61	ok
110-29	12	240	80	ok
110-30	7	140	170	ok
110-31	10	200	87	ok

Unit 111 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 55.7 ac, BAF 20 variable-radius cruise on a 300 ft grid, 27 plots, 300 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (SE 3.5 %)	Type	Sierran mixed conifer
Trees per acre	110	CWHR size / density	4 / D (cover 86 %)
QMD	19.2 in	SDI	296 (51 % of max 586)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	216 sq ft/ac 20.9 in	BA QMD	7 sq ft/ac 7.9 in
TPA	91	TPA	20
Volume	43.3 MBF/ac (7882 cu ft)	Volume	2.9 green tons/ac (96 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~154 sq ft/ac BA	Sampling error	7.1 % at 95 % (t = 2.06, n = 27)
Species (BA)	WF 71, PP 63, DF 36, IC 35, SP 18	Stratum standard	40 % - MEETS; 27 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	5	12.6	3.7	52	0.3
8-11.9	13	17.1	9.6	170	0.9
12-15.9	36	25.6	26.7	676	3.7
16-19.9	46	19.8	34.1	1051	5.8
20-23.9	49	14.1	36.3	1244	6.8
24-27.9	43	8.7	31.9	1188	6.5
28-31.9	38	5.9	28.1	1154	6.3
32-35.9	25	3.0	18.5	783	4.3
36-39.9	16	1.5	11.9	595	3.3
40-43.9	11	0.9	8.1	389	2.1
44-47.9	13	0.8	9.6	477	2.6
48-51.9	2	0.1	1.5	72	0.4
52-55.9	2	0.1	1.5	79	0.4
60-63.9	1	0.0	0.7	47	0.3
All	300	110.2	222.2	7977	43.9

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	96	34.7	71.1	19.4	2572	14.1
PP	85	32.9	63.0	18.7	2225	12.2
DF	48	16.5	35.6	19.9	1235	6.8
IC	47	15.8	34.8	20.1	1342	7.4
SP	24	10.4	17.8	17.7	604	3.3
All	300	110.2	222.2	19.2	7977	43.9

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 111: plots on the 300 ft grid (orange = QA flag)

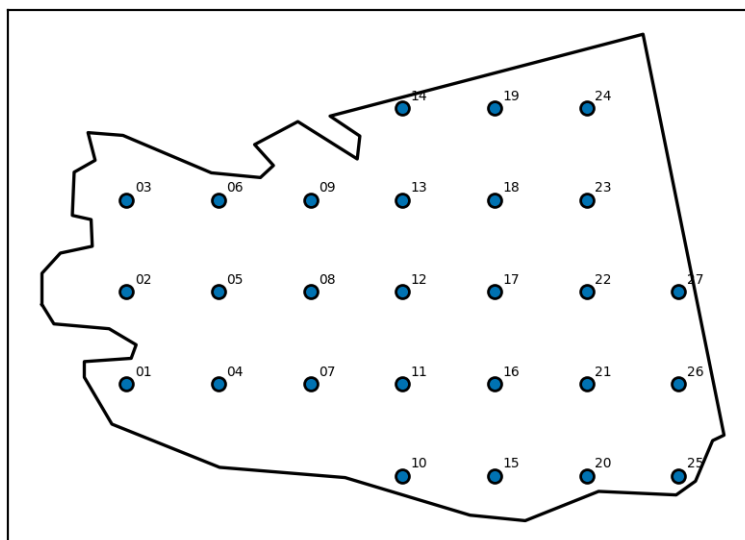


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
111-01	13	260	94	ok
111-02	11	220	49	ok
111-03	11	220	64	ok
111-04	9	180	60	ok
111-05	14	280	22	ok
111-06	10	200	25	ok
111-07	13	260	22	ok
111-08	9	180	30	ok
111-09	11	220	106	ok
111-10	8	160	99	ok
111-11	13	260	56	ok
111-12	14	280	65	ok
111-13	12	240	124	ok
111-14	10	200	122	ok
111-15	9	180	16	ok
111-16	11	220	86	ok
111-17	13	260	12	ok
111-18	13	260	32	ok
111-19	10	200	112	ok
111-20	9	180	127	ok
111-21	10	200	18	ok
111-22	12	240	62	ok
111-23	12	240	46	ok
111-24	12	240	62	ok
111-25	6	120	10	ok
111-26	11	220	75	ok
111-27	14	280	103	ok

Unit 112 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 80.5 ac, BAF 20 variable-radius cruise on a 300 ft grid, 40 plots, 428 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (SE 2.5 %)	Type	Sierran mixed conifer
Trees per acre	92	CWHR size / density	4 / D (cover 87 %)
QMD	20.6 in	SDI	279 (48 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	210 sq ft/ac 21.5 in	BA QMD	4 sq ft/ac 8.6 in
TPA	84	TPA	9
Volume	44.4 MBF/ac (8077 cu ft)	Volume	2.2 green tons/ac (72 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~156 sq ft/ac BA	Sampling error	5.1 % at 95 % (t = 2.02, n = 40)
Species (BA)	WF 66, PP 63, IC 36, DF 30, SP 19	Stratum standard	40 % - MEETS; 40 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	2	3.4	1.0	16	0.1
8-11.9	19	16.0	9.5	234	1.3
12-15.9	47	21.9	23.5	637	3.5
16-19.9	62	17.6	31.0	952	5.2
20-23.9	61	11.7	30.5	1068	5.9
24-27.9	69	9.5	34.5	1336	7.3
28-31.9	51	5.2	25.5	1066	5.9
32-35.9	42	3.4	21.0	926	5.1
36-39.9	28	1.8	14.0	680	3.7
40-43.9	12	0.6	6.0	296	1.6
44-47.9	12	0.5	6.0	307	1.7
48-51.9	9	0.3	4.5	247	1.4
52-55.9	9	0.3	4.5	248	1.4
56-59.9	1	0.0	0.5	26	0.1
60-63.9	4	0.1	2.0	111	0.6
All	428	92.5	214.0	8149	44.8

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	131	27.0	65.5	21.1	2522	13.9
PP	126	27.2	63.0	20.6	2383	13.1
IC	72	15.3	36.0	20.7	1415	7.8
DF	61	14.7	30.5	19.5	1130	6.2
SP	38	8.3	19.0	20.5	700	3.9
All	428	92.5	214.0	20.6	8149	44.8

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 112: plots on the 300 ft grid (orange = QA flag)

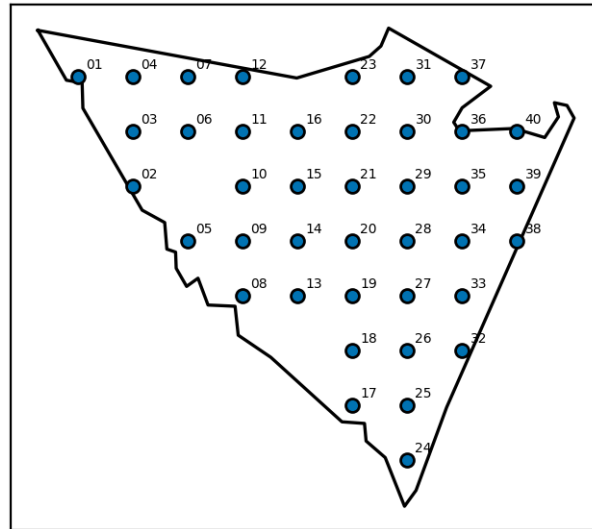


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
112-01	11	220	60	ok
112-02	11	220	24	ok
112-03	11	220	17	ok
112-04	12	240	30	ok
112-05	9	180	79	ok
112-06	12	240	147	ok
112-07	10	200	17	ok
112-08	5	100	112	ok
112-09	12	240	47	ok
112-10	11	220	43	ok
112-11	9	180	41	ok
112-12	11	220	129	ok
112-13	10	200	54	ok
112-14	13	260	21	ok
112-15	13	260	69	ok
112-16	13	260	72	ok
112-17	10	200	28	ok
112-18	12	240	105	ok
112-19	9	180	44	ok
112-20	11	220	98	ok
112-21	10	200	51	ok
112-22	13	260	33	ok
112-23	10	200	85	ok
112-24	13	260	96	ok
112-25	10	200	69	ok
112-26	12	240	61	ok
112-27	8	160	32	ok
112-28	10	200	44	ok
112-29	8	160	53	ok
112-30	11	220	26	ok
112-31	8	160	24	ok
112-32	12	240	77	ok
112-33	11	220	106	ok
112-34	11	220	68	ok
112-35	10	200	23	ok
112-36	13	260	186	ok
112-37	11	220	75	ok
112-38	9	180	72	ok
112-39	12	240	96	ok
112-40	11	220	114	ok

Unit 113 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 73.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 36 plots, 411 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	228 sq ft/ac (SE 2.0 %)	Type	Sierran mixed conifer
Trees per acre	70	CWHR size / density	5 / D (cover 95 %)
QMD	24.5 in	SDI	283 (49 % of max 573)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	228 sq ft/ac 24.5 in	BA QMD	0 sq ft/ac 0.0 in
TPA	70	TPA	0
Volume	52.8 MBF/ac (9594 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~162 sq ft/ac BA	Sampling error	4.1 % at 95 % (t = 2.03, n = 36)
Species (BA)	PP 70, WF 66, IC 36, DF 34, SP 23	Stratum standard	40 % - MEETS; 36 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	3	2.7	1.7	44	0.2
12-15.9	18	9.2	10.0	303	1.7
16-19.9	45	13.9	25.0	834	4.6
20-23.9	81	17.3	45.0	1670	9.2
24-27.9	70	10.6	38.9	1576	8.7
28-31.9	50	5.8	27.8	1162	6.4
32-35.9	49	4.3	27.2	1256	6.9
36-39.9	43	3.1	23.9	1185	6.5
40-43.9	21	1.2	11.7	606	3.3
44-47.9	13	0.6	7.2	385	2.1
48-51.9	5	0.2	2.8	158	0.9
52-55.9	5	0.2	2.8	152	0.8
56-59.9	4	0.1	2.2	127	0.7
60-63.9	4	0.1	2.2	135	0.7
All	411	69.5	228.3	9594	52.8

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	126	20.3	70.0	25.2	2965	16.3
WF	118	19.6	65.6	24.8	2815	15.5
IC	64	10.7	35.6	24.6	1484	8.2
DF	62	12.7	34.4	22.3	1368	7.5
SP	41	6.2	22.8	25.9	962	5.3
All	411	69.5	228.3	24.5	9594	52.8

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 113: plots on the 300 ft grid (orange = QA flag)

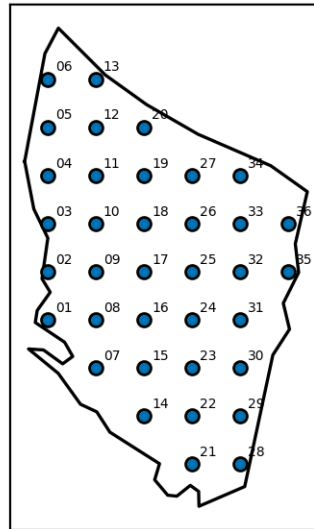


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
113-01	13	260	100	ok
113-02	14	280	69	ok
113-03	11	220	104	ok
113-04	11	220	101	ok
113-05	13	260	100	ok
113-06	13	260	155	ok
113-07	9	180	48	ok
113-08	13	260	67	ok
113-09	11	220	132	ok
113-10	12	240	78	ok
113-11	12	240	108	ok
113-12	11	220	78	ok
113-13	11	220	56	ok
113-14	12	240	86	ok
113-15	11	220	114	ok
113-16	13	260	91	ok
113-17	11	220	66	ok
113-18	12	240	38	ok
113-19	9	180	93	ok
113-20	10	200	60	ok
113-21	12	240	129	ok
113-22	10	200	93	ok
113-23	12	240	70	ok
113-24	12	240	88	ok
113-25	10	200	109	ok
113-26	10	200	102	ok
113-27	10	200	127	ok
113-28	10	200	67	ok
113-29	12	240	62	ok
113-30	12	240	24	ok
113-31	9	180	15	ok
113-32	11	220	41	ok
113-33	12	240	24	ok
113-34	15	300	46	ok
113-35	11	220	49	ok
113-36	11	220	32	ok

Unit 405 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 42.6 ac, BAF 20 variable-radius cruise on a 300 ft grid, 20 plots, 236 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	236 sq ft/ac (SE 2.3 %)	Type	Sierran mixed conifer
Trees per acre	61	CWHR size / density	5 / D (cover 95 %)
QMD	26.6 in	SDI	278 (47 % of max 593)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	235 sq ft/ac 27.0 in	BA QMD	1 sq ft/ac 9.5 in
TPA	59	TPA	2
Volume	63.2 MBF/ac (11484 cu ft)	Volume	0.6 green tons/ac (21 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~176 sq ft/ac BA	Sampling error	4.7 % at 95 % (t = 2.09, n = 20)
Species (BA)	WF 75, PP 57, IC 38, DF 37, SP 29	Stratum standard	40 % - MEETS; 20 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	3	5.3	3.0	78	0.4
12-15.9	10	9.4	10.0	284	1.6
16-19.9	20	11.0	20.0	680	3.7
20-23.9	22	8.6	22.0	850	4.7
24-27.9	34	9.3	34.0	1484	8.2
28-31.9	16	3.3	16.0	711	3.9
32-35.9	34	5.4	34.0	1630	9.0
36-39.9	23	3.0	23.0	1241	6.8
40-43.9	22	2.3	22.0	1251	6.9
44-47.9	14	1.2	14.0	834	4.6
48-51.9	9	0.7	9.0	535	2.9
52-55.9	12	0.8	12.0	753	4.1
56-59.9	6	0.3	6.0	398	2.2
60-63.9	11	0.6	11.0	777	4.3
All	236	61.0	236.0	11505	63.3

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	75	17.0	75.0	28.4	3684	20.3
PP	57	15.4	57.0	26.0	2821	15.5
IC	38	11.6	38.0	24.5	1880	10.3
DF	37	10.1	37.0	25.9	1688	9.3
SP	29	6.9	29.0	27.7	1431	7.9
All	236	61.0	236.0	26.6	11505	63.3

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 405: plots on the 300 ft grid (orange = QA flag)

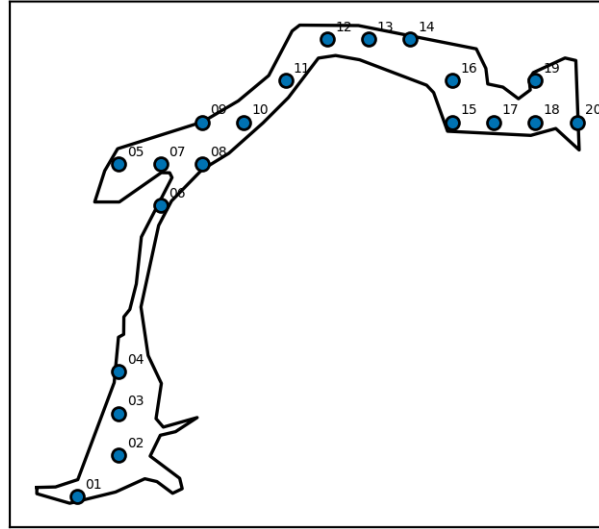


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
405-01	12	240	233	ok
405-02	12	240	267	ok
405-03	10	200	225	ok
405-04	12	240	257	ok
405-05	11	220	336	ok
405-06	11	220	203	ok
405-07	12	240	218	ok
405-08	12	240	140	ok
405-09	11	220	122	ok
405-10	12	240	239	ok
405-11	12	240	199	ok
405-12	13	260	182	ok
405-13	12	240	217	ok
405-14	15	300	247	ok
405-15	10	200	114	ok
405-16	14	280	162	ok
405-17	11	220	174	ok
405-18	11	220	174	ok
405-19	12	240	107	ok
405-20	11	220	150	ok

Unit 114 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 116.3 ac, BAF 20 variable-radius cruise on a 300 ft grid, 56 plots, 593 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	212 sq ft/ac (SE 2.1 %)	Type	Sierran mixed conifer
Trees per acre	93	CWHR size / density	4 / D (cover 85 %)
QMD	20.5 in	SDI	276 (48 % of max 573)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	208 sq ft/ac 21.6 in	BA QMD	4 sq ft/ac 7.9 in
TPA	81	TPA	11
Volume	43.4 MBF/ac (7899 cu ft)	Volume	2.1 green tons/ac (69 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~154 sq ft/ac BA	Sampling error	4.3 % at 95 % (t = 2.00, n = 56)
Species (BA)	PP 65, WF 61, IC 33, DF 30, SP 22	Stratum standard	40 % - MEETS; 56 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	5	7.0	1.8	28	0.2
8-11.9	23	13.4	8.2	173	1.0
12-15.9	70	22.5	25.0	672	3.7
16-19.9	78	15.8	27.9	854	4.7
20-23.9	103	14.3	36.8	1254	6.9
24-27.9	70	7.0	25.0	930	5.1
28-31.9	79	5.7	28.2	1189	6.5
32-35.9	59	3.4	21.1	951	5.2
36-39.9	30	1.3	10.7	501	2.8
40-43.9	25	0.9	8.9	433	2.4
44-47.9	21	0.7	7.5	389	2.1
48-51.9	9	0.2	3.2	168	0.9
52-55.9	5	0.1	1.8	98	0.5
56-59.9	7	0.1	2.5	128	0.7
60-63.9	9	0.2	3.2	200	1.1
All	593	92.8	211.8	7968	43.8

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	183	32.0	65.4	19.4	2398	13.2
WF	172	25.3	61.4	21.1	2370	13.0
IC	92	13.8	32.9	20.9	1233	6.8
DF	84	12.6	30.0	20.9	1153	6.3
SP	62	9.1	22.1	21.1	815	4.5
All	593	92.8	211.8	20.5	7968	43.8

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 114: plots on the 300 ft grid (orange = QA flag)

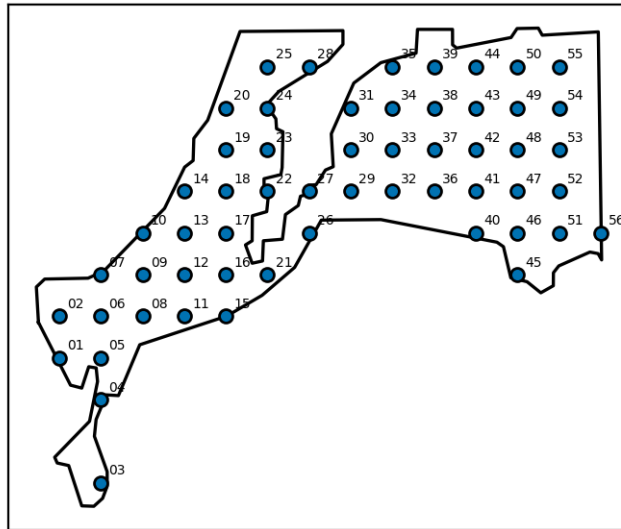


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
114-01	12	240	120	ok
114-02	9	180	15	ok
114-03	8	160	82	ok
114-04	12	240	117	ok
114-05	9	180	39	ok
114-06	11	220	60	ok
114-07	10	200	87	ok
114-08	14	280	109	ok
114-09	11	220	60	ok
114-10	12	240	100	ok
114-11	12	240	62	ok
114-12	12	240	126	ok
114-13	9	180	41	ok
114-14	11	220	33	ok
114-15	10	200	122	ok
114-16	11	220	63	ok
114-17	11	220	45	ok
114-18	10	200	15	ok
114-19	13	260	23	ok
114-20	12	240	76	ok
114-21	12	240	24	ok
114-22	9	180	54	ok
114-23	12	240	36	ok
114-24	8	160	16	ok
114-25	12	240	132	ok
114-26	9	180	27	ok
114-27	9	180	21	ok
114-28	11	220	420	ok
114-29	8	160	26	ok
114-30	12	240	28	ok
114-31	10	200	56	ok
114-32	10	200	75	ok
114-33	12	240	37	ok
114-34	11	220	55	ok
114-35	8	160	52	ok
114-36	6	120	28	ok
114-37	13	260	42	ok
114-38	12	240	155	ok
114-39	11	220	22	ok
114-40	11	220	75	ok
114-41	10	200	62	ok
114-42	11	220	39	ok
114-43	10	200	41	ok
114-44	10	200	42	ok
114-45	14	280	143	ok
114-46	10	200	75	ok
114-47	12	240	56	ok
114-48	8	160	39	ok
114-49	7	140	34	ok
114-50	10	200	31	ok
114-51	10	200	18	ok

Plot	Trees in	BA sq ft/ac	Slope %	Status
114-52	10	200	24	ok
114-53	12	240	38	ok
114-54	10	200	26	ok
114-55	11	220	23	ok
114-56	13	260	19	ok

Unit 115 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 84.0 ac, BAF 20 variable-radius cruise on a 300 ft grid, 39 plots, 444 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	228 sq ft/ac (SE 2.0 %)	Type	Sierran mixed conifer
Trees per acre	87	CWHR size / density	4 / D (cover 92 %)
QMD	21.9 in	SDI	292 (51 % of max 578)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	227 sq ft/ac 22.2 in	BA QMD	1 sq ft/ac 8.2 in
TPA	85	TPA	3
Volume	47.5 MBF/ac (8636 cu ft)	Volume	0.6 green tons/ac (21 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~158 sq ft/ac BA	Sampling error	4.0 % at 95 % (t = 2.02, n = 39)
Species (BA)	WF 68, PP 67, IC 34, DF 33, SP 26	Stratum standard	40 % - MEETS; 39 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	1.8	0.5	10	0.1
8-11.9	11	8.8	5.6	133	0.7
12-15.9	43	20.2	22.1	615	3.4
16-19.9	64	18.7	32.8	971	5.3
20-23.9	73	14.2	37.4	1308	7.2
24-27.9	68	9.8	34.9	1342	7.4
28-31.9	51	5.4	26.2	1084	6.0
32-35.9	48	3.9	24.6	1083	6.0
36-39.9	31	2.1	15.9	731	4.0
40-43.9	22	1.2	11.3	521	2.9
44-47.9	15	0.7	7.7	381	2.1
48-51.9	9	0.3	4.6	249	1.4
52-55.9	5	0.2	2.6	132	0.7
60-63.9	3	0.1	1.5	97	0.5
All	444	87.3	227.7	8657	47.6

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	132	24.4	67.7	22.6	2613	14.4
PP	130	24.9	66.7	22.2	2557	14.1
IC	66	14.5	33.8	20.7	1223	6.7
DF	65	12.4	33.3	22.2	1317	7.2
SP	51	11.2	26.2	20.7	948	5.2
All	444	87.3	227.7	21.9	8657	47.6

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 115: plots on the 300 ft grid (orange = QA flag)

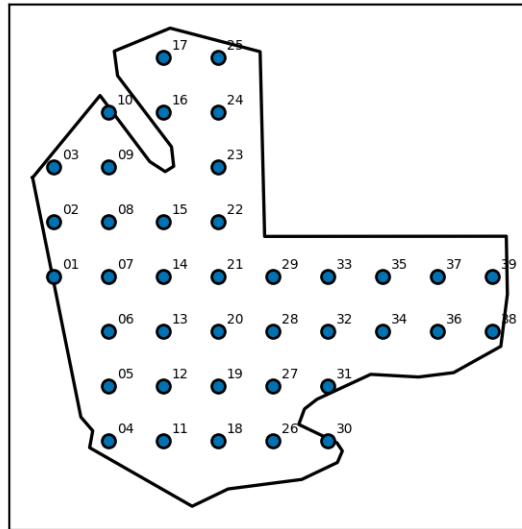


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
115-01	13	260	84	ok
115-02	10	200	36	ok
115-03	8	160	36	ok
115-04	12	240	43	ok
115-05	12	240	101	ok
115-06	12	240	36	ok
115-07	11	220	144	ok
115-08	11	220	38	ok
115-09	11	220	43	ok
115-10	9	180	91	ok
115-11	12	240	24	ok
115-12	12	240	29	ok
115-13	10	200	52	ok
115-14	10	200	30	ok
115-15	12	240	29	ok
115-16	9	180	104	ok
115-17	11	220	96	ok
115-18	11	220	25	ok
115-19	8	160	24	ok
115-20	11	220	38	ok
115-21	11	220	81	ok
115-22	14	280	41	ok
115-23	12	240	22	ok
115-24	12	240	53	ok
115-25	12	240	65	ok
115-26	10	200	24	ok
115-27	12	240	20	ok
115-28	12	240	38	ok
115-29	12	240	16	ok
115-30	11	220	11	ok
115-31	12	240	11	ok
115-32	12	240	36	ok
115-33	12	240	11	ok
115-34	12	240	37	ok
115-35	13	260	12	ok
115-36	15	300	14	ok
115-37	11	220	37	ok
115-38	12	240	21	ok
115-39	12	240	25	ok

Unit 116 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 89.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 44 plots, 507 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	230 sq ft/ac (SE 1.5 %)	Type	Sierran mixed conifer
Trees per acre	70	CWHR size / density	5 / D (cover 94 %)
QMD	24.6 in	SDI	283 (49 % of max 573)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	230 sq ft/ac 24.9 in	BA QMD	1 sq ft/ac 9.4 in
TPA	68	TPA	2
Volume	53.6 MBF/ac (9740 cu ft)	Volume	0.7 green tons/ac (22 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~163 sq ft/ac BA	Sampling error	3.1 % at 95 % (t = 2.02, n = 44)
Species (BA)	PP 65, WF 62, DF 42, IC 36, SP 25	Stratum standard	40 % - MEETS; 44 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	6	4.9	2.7	64	0.4
12-15.9	27	10.5	12.3	362	2.0
16-19.9	57	14.7	25.9	847	4.7
20-23.9	71	12.3	32.3	1154	6.3
24-27.9	86	10.7	39.1	1571	8.6
28-31.9	70	6.4	31.8	1407	7.7
32-35.9	58	4.3	26.4	1212	6.7
36-39.9	44	2.6	20.0	985	5.4
40-43.9	29	1.4	13.2	686	3.8
44-47.9	15	0.6	6.8	352	1.9
48-51.9	21	0.7	9.5	520	2.9
52-55.9	10	0.3	4.5	253	1.4
56-59.9	5	0.1	2.3	132	0.7
60-63.9	8	0.2	3.6	218	1.2
All	507	69.7	230.5	9763	53.7

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	144	17.6	65.5	26.1	2802	15.4
WF	136	19.6	61.8	24.0	2622	14.4
DF	93	13.1	42.3	24.4	1766	9.7
IC	79	11.8	35.9	23.6	1482	8.2
SP	55	7.6	25.0	24.6	1089	6.0
All	507	69.7	230.5	24.6	9763	53.7

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 116: plots on the 300 ft grid (orange = QA flag)

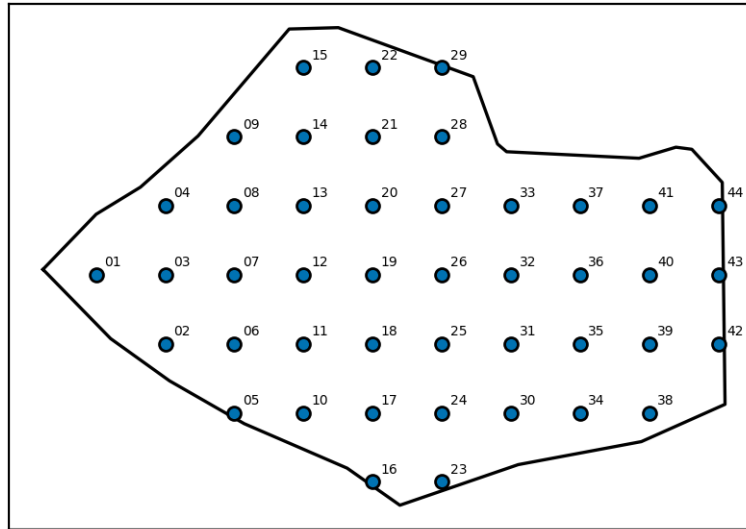


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
116-01	10	200	57	ok
116-02	12	240	57	ok
116-03	11	220	36	ok
116-04	10	200	72	ok
116-05	11	220	69	ok
116-06	12	240	67	ok
116-07	10	200	98	ok
116-08	10	200	47	ok
116-09	12	240	82	ok
116-10	11	220	63	ok
116-11	11	220	46	ok
116-12	14	280	46	ok
116-13	11	220	22	ok
116-14	11	220	21	ok
116-15	12	240	17	ok
116-16	10	200	53	ok
116-17	13	260	37	ok
116-18	11	220	52	ok
116-19	12	240	70	ok
116-20	12	240	31	ok
116-21	11	220	18	ok
116-22	10	200	17	ok
116-23	11	220	30	ok
116-24	13	260	38	ok
116-25	11	220	83	ok
116-26	11	220	68	ok
116-27	13	260	33	ok
116-28	13	260	200	ok
116-29	13	260	96	ok
116-30	11	220	94	ok
116-31	12	240	60	ok
116-32	14	280	39	ok
116-33	11	220	37	ok
116-34	10	200	18	ok
116-35	11	220	41	ok
116-36	11	220	32	ok
116-37	12	240	74	ok
116-38	13	260	28	ok
116-39	12	240	32	ok
116-40	13	260	39	ok
116-41	9	180	10	ok
116-42	11	220	44	ok
116-43	13	260	9	ok
116-44	12	240	53	ok

Unit 117 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 82.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 44 plots, 490 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	223 sq ft/ac (SE 2.0 %)	Type	Sierran mixed conifer
Trees per acre	89	CWHR size / density	4 / D (cover 88 %)
QMD	21.4 in	SDI	286 (49 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	219 sq ft/ac 22.4 in	BA QMD	4 sq ft/ac 8.7 in
TPA	80	TPA	9
Volume	46.8 MBF/ac (8507 cu ft)	Volume	2.0 green tons/ac (67 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~158 sq ft/ac BA	Sampling error	4.0 % at 95 % (t = 2.02, n = 44)
Species (BA)	WF 69, PP 66, IC 36, DF 33, SP 18	Stratum standard	40 % - MEETS; 44 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	1.4	0.5	7	0.0
8-11.9	20	16.0	9.1	194	1.1
12-15.9	40	17.3	18.2	482	2.6
16-19.9	67	17.1	30.5	920	5.1
20-23.9	83	14.2	37.7	1325	7.3
24-27.9	71	8.9	32.3	1247	6.9
28-31.9	66	6.2	30.0	1264	7.0
32-35.9	50	3.6	22.7	1007	5.5
36-39.9	31	1.8	14.1	657	3.6
40-43.9	21	1.0	9.5	483	2.7
44-47.9	19	0.7	8.6	450	2.5
48-51.9	9	0.3	4.1	225	1.2
52-55.9	7	0.2	3.2	176	1.0
56-59.9	3	0.1	1.4	85	0.5
60-63.9	2	0.0	0.9	52	0.3
All	490	89.0	222.7	8574	47.2

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	151	26.6	68.6	21.8	2674	14.7
PP	146	28.4	66.4	20.7	2531	13.9
IC	80	15.3	36.4	20.8	1412	7.8
DF	73	12.5	33.2	22.0	1251	6.9
SP	40	6.1	18.2	23.3	706	3.9
All	490	89.0	222.7	21.4	8574	47.2

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 117: plots on the 300 ft grid (orange = QA flag)

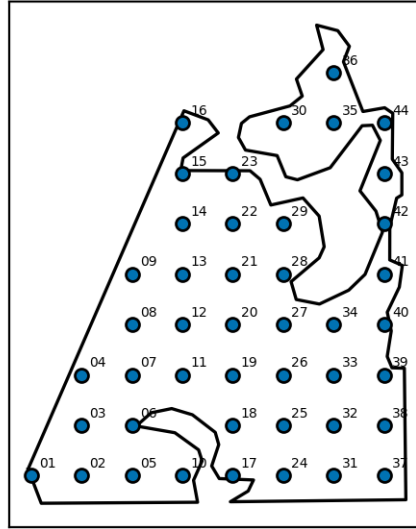


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
117-01	11	220	17	ok
117-02	12	240	201	ok
117-03	13	260	85	ok
117-04	13	260	35	ok
117-05	10	200	94	ok
117-06	10	200	170	ok
117-07	11	220	36	ok
117-08	12	240	92	ok
117-09	9	180	64	ok
117-10	11	220	57	ok
117-11	12	240	60	ok
117-12	11	220	45	ok
117-13	13	260	9	ok
117-14	12	240	58	ok
117-15	12	240	115	ok
117-16	10	200	65	ok
117-17	9	180	116	ok
117-18	11	220	78	ok
117-19	11	220	104	ok
117-20	12	240	111	ok
117-21	12	240	28	ok
117-22	13	260	25	ok
117-23	10	200	98	ok
117-24	11	220	130	ok
117-25	6	120	48	ok
117-26	11	220	118	ok
117-27	13	260	130	ok
117-28	12	240	98	ok
117-29	12	240	95	ok
117-30	13	260	37	ok
117-31	10	200	139	ok
117-32	10	200	63	ok
117-33	12	240	95	ok
117-34	11	220	15	ok
117-35	12	240	92	ok
117-36	11	220	87	ok
117-37	11	220	200	ok
117-38	13	260	54	ok
117-39	12	240	80	ok
117-40	12	240	183	ok
117-41	9	180	50	ok
117-42	9	180	118	ok
117-43	11	220	85	ok
117-44	9	180	69	ok

Unit 118 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 70.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 34 plots, 374 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	220 sq ft/ac (SE 3.2 %)	Type	Sierran mixed conifer
Trees per acre	76	CWHR size / density	4 / D (cover 91 %)
QMD	23.1 in	SDI	278 (47 % of max 595)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	219 sq ft/ac 23.4 in	BA QMD	1 sq ft/ac 8.7 in
TPA	73	TPA	3
Volume	48.5 MBF/ac (8819 cu ft)	Volume	0.8 green tons/ac (28 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~165 sq ft/ac BA	Sampling error	6.4 % at 95 % (t = 2.03, n = 34)
Species (BA)	WF 76, PP 59, DF 38, IC 35, SP 13	Stratum standard	40 % - MEETS; 34 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	7	6.9	4.1	113	0.6
12-15.9	29	14.9	17.1	501	2.8
16-19.9	42	13.8	24.7	831	4.6
20-23.9	73	16.5	42.9	1527	8.4
24-27.9	67	10.7	39.4	1579	8.7
28-31.9	43	5.2	25.3	1076	5.9
32-35.9	35	3.4	20.6	892	4.9
36-39.9	28	2.1	16.5	774	4.3
40-43.9	11	0.7	6.5	328	1.8
44-47.9	19	1.0	11.2	569	3.1
48-51.9	6	0.3	3.5	179	1.0
52-55.9	3	0.1	1.8	96	0.5
56-59.9	4	0.1	2.4	132	0.7
60-63.9	7	0.2	4.1	250	1.4
All	374	75.9	220.0	8847	48.7

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	129	29.5	75.9	21.7	2983	16.4
PP	100	21.6	58.8	22.4	2351	12.9
DF	64	11.7	37.6	24.3	1531	8.4
IC	59	9.4	34.7	26.0	1456	8.0
SP	22	3.7	12.9	25.5	526	2.9
All	374	75.9	220.0	23.1	8847	48.7

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 118: plots on the 300 ft grid (orange = QA flag)

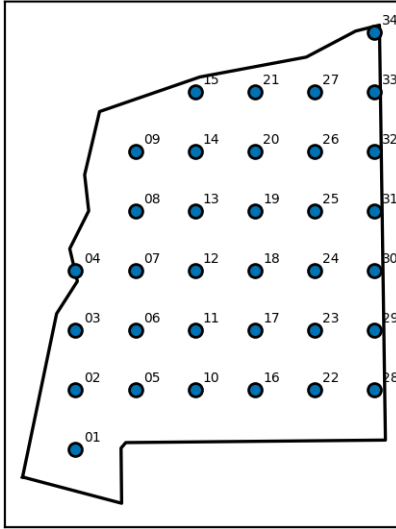


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
118-01	12	240	34	ok
118-02	10	200	32	ok
118-03	12	240	59	ok
118-04	9	180	17	ok
118-05	15	300	49	ok
118-06	12	240	59	ok
118-07	11	220	51	ok
118-08	11	220	88	ok
118-09	10	200	58	ok
118-10	5	100	36	ok
118-11	10	200	63	ok
118-12	13	260	78	ok
118-13	12	240	65	ok
118-14	12	240	47	ok
118-15	11	220	70	ok
118-16	8	160	51	ok
118-17	12	240	61	ok
118-18	11	220	32	ok
118-19	11	220	30	ok
118-20	7	140	132	ok
118-21	13	260	72	ok
118-22	8	160	66	ok
118-23	11	220	36	ok
118-24	13	260	70	ok
118-25	10	200	42	ok
118-26	11	220	53	ok
118-27	10	200	23	ok
118-28	15	300	39	ok
118-29	13	260	18	ok
118-30	11	220	28	ok
118-31	12	240	25	ok
118-32	10	200	43	ok
118-33	12	240	59	ok
118-34	11	220	35	ok

Unit 406 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 35.1 ac, BAF 20 variable-radius cruise on a 300 ft grid, 17 plots, 205 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	241 sq ft/ac (SE 2.3 %)	Type	Sierran mixed conifer
Trees per acre	71	CWHR size / density	5 / D (cover 93 %)
QMD	24.9 in	SDI	297 (50 % of max 598)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	241 sq ft/ac 24.9 in	BA QMD	0 sq ft/ac 0.0 in
TPA	71	TPA	0
Volume	56.8 MBF/ac (10335 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~170 sq ft/ac BA	Sampling error	4.9 % at 95 % (t = 2.12, n = 17)
Species (BA)	WF 82, PP 60, IC 42, DF 34, SP 22	Stratum standard	40 % - MEETS; 17 plots vs 20-plot practice: SHORT by 3

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	1	1.6	1.2	32	0.2
12-15.9	12	12.2	14.1	432	2.4
16-19.9	17	11.5	20.0	688	3.8
20-23.9	41	18.1	48.2	1773	9.8
24-27.9	29	9.6	34.1	1376	7.6
28-31.9	34	8.2	40.0	1727	9.5
32-35.9	21	4.0	24.7	1153	6.3
36-39.9	19	2.9	22.4	1106	6.1
40-43.9	13	1.6	15.3	813	4.5
44-47.9	7	0.7	8.2	468	2.6
48-51.9	5	0.4	5.9	336	1.9
52-55.9	1	0.1	1.2	80	0.4
56-59.9	3	0.2	3.5	211	1.2
60-63.9	2	0.1	2.4	140	0.8
All	205	71.2	241.2	10335	56.8

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	70	21.1	82.4	26.7	3730	20.5
PP	51	21.5	60.0	22.6	2396	13.2
IC	36	12.1	42.4	25.4	1856	10.2
DF	29	10.2	34.1	24.8	1395	7.7
SP	19	6.3	22.4	25.5	958	5.3
All	205	71.2	241.2	24.9	10335	56.8

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 406: plots on the 300 ft grid (orange = QA flag)

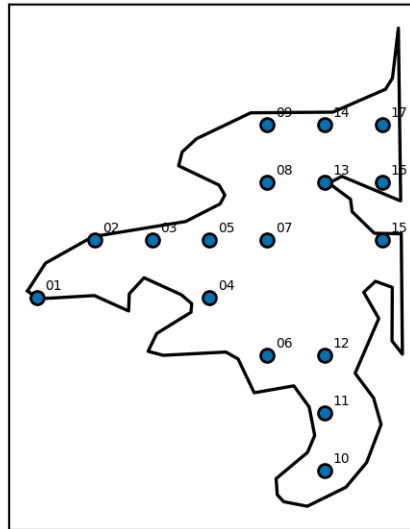


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted; 3 more plots would meet the 20-plot practice

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
406-01	13	260	186	ok
406-02	12	240	171	ok
406-03	12	240	259	ok
406-04	13	260	212	ok
406-05	11	220	177	ok
406-06	14	280	114	ok
406-07	12	240	260	ok
406-08	11	220	206	ok
406-09	10	200	185	ok
406-10	11	220	153	ok
406-11	12	240	212	ok
406-12	11	220	97	ok
406-13	12	240	147	ok
406-14	13	260	170	ok
406-15	11	220	166	ok
406-16	13	260	139	ok
406-17	14	280	152	ok